

CONSTRUCTION TECHNOLOGY

Construction technology has always played a central role in society. It is a multifaceted industry in which projects can result in products, systems, processes, or services and encompass various aspects of production, repair, and maintenance. In construction technology courses, students will learn about designing, constructing, and maintaining a variety of buildings and structures, and will gain experience with the tools, equipment, and processes commonly used in the field. Students will also learn about health and safety standards in the construction industry, building codes and regulations, and employment opportunities and careers in the various sectors of the industry, including electricity, carpentry, masonry, heating and cooling, and plumbing.

The construction technology courses initially focus on residential and light construction systems related to residential buildings and structures, and progress to more advanced residential and light commercial construction. The woodworking courses allow students to concentrate on furniture building or cabinet making.

The list of approved emphasis areas for construction technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Construction Engineering Technology, Grade 11

College Preparation

TCJ3C

This course focuses on the development of knowledge and skills related to residential construction. Students will gain hands-on experience using a variety of construction materials, processes, tools, and equipment; learn about building design and planning construction projects; create and interpret working drawings and sections; and learn how the Ontario Building Code and other regulations and standards apply to construction projects. Students will also develop an awareness of environmental and societal issues related to construction technology, and will explore career opportunities in the field.

Prerequisite: None

A. CONSTRUCTION TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of construction materials, processes, and components;
- A2.** describe the building codes, regulations, and standards that govern construction projects;
- A3.** demonstrate an understanding of the systems in a residential building;
- A4.** demonstrate an understanding of design considerations for residential buildings;
- A5.** use construction terminology correctly.

SPECIFIC EXPECTATIONS

A1. Materials, Processes, and Components

By the end of this course, students will:

- A1.1** identify and describe the uses and features of construction materials (*e.g., lumber, concrete, oriented-strand board [OSB], adhesives, finishes, hardware*), tools (*e.g., hammers, pliers, hand saws, circular saws, drills*), and equipment (*e.g., generators, scaffolds*);
- A1.2** identify and describe the properties (*e.g., physical, mechanical, thermal*) of common natural and manufactured building materials (*e.g., natural wood, engineered wood, steel framing*);
- A1.3** identify and describe the processes used to produce a variety of construction materials (*e.g., concrete, masonry, metal products, wood products*);
- A1.4** identify and describe various residential construction processes (*e.g., slip forming, insulated concrete forming, platform framing, balloon framing, assembling modular cabinetry*);
- A1.5** identify and describe structural and non-structural components of buildings (*e.g., structural: footings, foundations, floors, bearing walls, columns, beams, lintels, trusses, rafters; non-structural: trim, siding, flooring*).

A2. Building Codes, Regulations, and Standards

By the end of this course, students will:

- A2.1** describe the purpose of building codes (*e.g., Ontario Building Code, Ontario Electrical Safety Code, Ontario Fire and Plumbing codes*);

- A2.2** identify and describe regulations and/or by-laws that apply to the design and construction of residential buildings (*e.g., municipal by-laws; zoning regulations; official plans; requirements for setbacks, easements, and barrier-free access*);

- A2.3** identify and describe standards that apply to residential construction projects (*e.g., standards from the Canadian Standards Association [CSA], Underwriters Laboratories of Canada [ULC], and the Workplace Safety and Insurance Board [WSIB]*);

- A2.4** identify permits and inspections required for residential construction projects (*e.g., building, plumbing, electrical*);

- A2.5** describe how standards for the quality and processing of materials (*e.g., lumber grades, CSA specifications for pipe and wire*) affect the design of a building;

- A2.6** identify organizations that promote sustainable building practices, and related standards (*e.g., Canada Green Building Council, Leadership in Energy and Environmental Design [LEED®] certification standards*).

A3. Building Systems

By the end of this course, students will:

- A3.1** identify and describe the systems in a typical building (*e.g., foundation, framing, electrical, plumbing, roof*);
- A3.2** describe the relationship between structural systems and architectural elements in construction (*e.g., cladding on an exterior support pier*);

A3.3 identify and describe modifications to building systems that would improve the quality and value of the building (e.g., *upgraded ventilation, insulation, air barrier*).

A4. Design Considerations

By the end of this course, students will:

A4.1 describe the features of different types of houses (e.g., *one-storey, two-storey, split-level, duplex*);

A4.2 identify a variety of building types (e.g., *residential, institutional*), architectural styles (e.g., *Colonial, Tudor, Victorian, Aboriginal*), and construction engineering features (e.g., *post and beam, keystone, steel supporting column*);

A4.3 identify components of an environmentally friendly house (e.g., *solar water heater, energy-efficient heating and cooling systems, recycled building materials*);

A4.4 describe human factors (e.g., *function, ergonomics, aesthetics*) to consider when planning aspects of a house (e.g., *size, appearance, traffic flow, means of egress, use of non-allergenic materials*);

A4.5 identify and describe the reasons for selecting a particular structure and materials for a construction project (e.g., *structure: wood versus steel beams; materials: asphalt shingles versus steel roofing*);

A4.6 identify factors affecting the design of a foundation (e.g., *drainage, soil type, load, frost penetration*);

A4.7 identify the weather-related loads and stresses that a building must be designed to withstand (e.g., *force of wind, snow load on roofs, expansion and contraction due to changes in temperature and humidity*).

A5. Terminology

By the end of this course, students will:

A5.1 use correct terminology to identify and describe materials, supplies, and structural components (e.g., *materials: plywood, concrete, polyvinyl chloride [PVC] pipe, wire and cable; supplies: P-trap, ground fault circuit interrupter; structural components: footing, stud, lintel, plenum, valance, gable*);

A5.2 use correct terminology to describe construction processes and techniques (e.g., *processes: joining, levelling, squaring, making plumb; techniques: toe-nailing, shimming*);

A5.3 use correct terminology to identify and describe construction tools and equipment (e.g., *tools: framing hammer, level, wire stripper, tri-square, power nailer; equipment: air compressor, cement mixer*).

B. DESIGN, LAYOUT, AND PLANNING SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** apply a design process and other problem-solving processes and techniques as appropriate to develop solutions for construction problems or challenges;
- B2.** create and use working drawings for a variety of residential construction projects;
- B3.** determine, use, and communicate accurate technical data in the design of construction projects;
- B4.** use the mathematical skills required in designing, laying out, and preparing estimates for construction projects.

SPECIFIC EXPECTATIONS

B1. Design and Planning

By the end of this course, students will:

- B1.1** identify the steps of a design process (see pp. 22–23), and describe how the design process is used in the construction industry (*e.g., preparing floor plans, designing prefabricated components; in urban planning*);
- B1.2** use appropriate problem-solving processes and techniques (see pp. 21–23) to address challenges related to various residential construction projects (*e.g., house, cottage, shed, renovation*);
- B1.3** use models, prototypes, and/or sketches to aid in improving the design of construction projects;
- B1.4** apply design principles (*e.g., scale, proportion, contrast*) when designing systems in construction projects (*e.g., placement of electrical and plumbing fixtures*).

B2. Technical Drawings

By the end of this course, students will:

- B2.1** identify the symbols, abbreviations, hatchings, and other conventions used in various types of drawings of residential buildings (*e.g., layout, foundation, floor plan, elevation, cross-section, detail, electrical system, mechanical system*);
- B2.2** interpret technical drawings and specifications to accurately plan and lay out residential construction projects (*e.g., determine dimensions,*

materials required, windows and door types, and locations of electrical devices and plumbing fixtures);

- B2.3** modify and/or create hand-drafted and/or computer-assisted technical drawings (*e.g., orthographic, isometric, plan, elevation, plumbing, cross-section*), using appropriate metric and/or imperial units.

B3. Using Technical Data

By the end of this course, students will:

- B3.1** use appropriate sources to determine the technical requirements for residential construction projects (*e.g., technical charts and tables, engineering reports, building codes and regulations, standards, municipal by-laws*);
- B3.2** identify and describe the types of loads (*e.g., live, dead*) that building codes require residential structural members (*e.g., footings, foundations, floors, bearing walls, roofs, columns, beams, lintels*) to withstand;
- B3.3** communicate accurate technical information verbally and graphically (*e.g., through written instructions, sketches, and/or detail drawings*).

B4. Mathematical Skills

By the end of this course, students will:

- B4.1** determine dimensions and lay out construction projects using mathematical principles and formulas (*e.g., Pythagorean theorem, volume and area formulas*);

B4.2 use the tables in Part 9 of the Ontario Building Code to determine the required sizes, spacing, and numbers of structural components (*e.g., floor joists, lintels*) for a construction project;

B4.3 prepare detailed, accurate estimates of quantities and costs of materials for construction projects, using appropriate metric and/or imperial units (*e.g., metres, square metres, litres, board feet, linear feet, square feet, cubic yards*);

B4.4 determine lengths and diameters of fastening devices needed to assemble various construction projects (*e.g., lengths and gauges of screws, nails, and staples; diameters of dowels*), using appropriate metric and/or imperial units.

C. FABRICATION, ASSEMBLY, AND FINISHING SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate appropriate technical skills, including the safe use of construction tools, equipment, and materials;
- C2.** demonstrate safe and accurate building techniques;
- C3.** correctly apply various finishes to complete residential construction projects.

SPECIFIC EXPECTATIONS

C1. Technical Skills

By the end of this course, students will:

- C1.1** demonstrate safe work practices when using hand tools, power tools, equipment, and materials;
- C1.2** correctly use metric and imperial measuring tools and equipment in a variety of construction projects;
- C1.3** lay out construction projects using a variety of tools and equipment (*e.g., framing square, laser level, string line, plumb bob*).

C2. Fabrication and Assembly

By the end of this course, students will:

- C2.1** safely construct projects in accordance with the specifications for the project (*e.g., technical drawings, lists of specified materials and fixtures*);
- C2.2** install various systems of a construction project (*e.g., structural, electrical, plumbing, heating/ventilation/air-conditioning, cabinetry*) in accordance with codes, regulations, and standards (*e.g., building code requirements for size and spacing of joists or beams, Ontario Electrical Safety Code requirements for wiring and fixtures*);
- C2.3** describe modifications to improve residential construction projects, and make these modifications where appropriate.

C3. Finishing

By the end of this course, students will:

- C3.1** identify and correctly apply appropriate exterior finishing materials (*e.g., siding, stucco, deck stain*);
- C3.2** identify and correctly apply appropriate interior finishing materials (*e.g., paint, wall-paper, trim, casing*).

D. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of the environmental effects of construction projects, and ways of reducing harmful effects;
- D2.** describe how society and the construction industry affect each other.

SPECIFIC EXPECTATIONS

D1. Technology and the Environment

By the end of this course, students will:

- D1.1** plan projects and processes to minimize waste (*e.g., use efficient cutting patterns, reuse and recycle leftover materials*);
- D1.2** identify and describe environmentally friendly building practices (*e.g., high-efficiency heating and cooling, renewable energy technologies, reuse of grey water, use of materials produced from sustainable resources*);
- D1.3** describe the environmental effects of using natural and manufactured construction materials (*e.g., energy use, release of toxic chemicals, disposal of manufacturing and construction waste, effects on water supply and quality*);
- D1.4** describe ways of reducing and/or managing energy consumption in the home (*e.g., smart meters, energy-efficient lighting, timers, heat-recovery ventilators*);
- D1.5** describe the life cycle of a construction product (*e.g., manufacture; installation; reuse, recycling, or disposal*).

D2. Technology and Society

By the end of this course, students will:

- D2.1** describe the economic and social effects of the construction industry (*e.g., land use, creation of primary and secondary jobs, transportation of equipment and materials; encroachment on Aboriginal lands*);
- D2.2** describe factors affecting the quality of life of the occupants of residential buildings (*e.g., air quality; allergens; access to schools, transit, and health care*);
- D2.3** describe how societal and client needs (*e.g., budget constraints, barrier-free access, energy efficiency, cultural practices*) affect construction projects.

E. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1.** demonstrate an understanding of and comply with health and safety regulations and practices specific to the construction industry;
- E2.** describe career opportunities in the construction industry, and the importance of lifelong learning for these careers.

SPECIFIC EXPECTATIONS

E1. Health and Safety

By the end of this course, students will:

- E1.1** describe hazards related to construction materials, processes, tools, and equipment (*e.g., toxic or flammable fumes from solvents, paints, varnishes, and gasoline; explosion or burns from propane; lung damage from silica; tripping or falls in unfinished buildings; shock from damaged power tools or electrical equipment*), and the precautions that should be taken to avoid these hazards;
- E1.2** outline and comply with health and safety legislation and practices for the construction industry (*e.g., Workplace Safety and Insurance Board [WSIB] regulations, provincial labour legislation, Ontario Building Code, local by-laws*);
- E1.3** use, handle, and store materials in accordance with Workplace Hazardous Materials Information System (WHMIS) guidelines;
- E1.4** describe the rights and responsibilities of employees (*e.g., the right to know, the right to refuse, the right to participate, as outlined in the Occupational Health and Safety Act*);
- E1.5** use protective clothing, gear, and equipment appropriately (*e.g., dust mask, hard hat, safety glasses, safety harness*).
- E2. Career Opportunities**
- By the end of this course, students will:
- E2.1** identify careers in construction technology (*e.g., civil or construction engineer, architectural draftsman, renovation technician or technologist, indigenous environmental technician, project manager*), and describe the education and training required for these careers;
- E2.2** describe the importance of lifelong learning for careers in the construction industry;
- E2.3** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the construction industry (*e.g., mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);
- E2.4** demonstrate an understanding of and apply the Essential Skills that are important for success in the construction industry, as identified in the Ontario Skills Passport (*e.g., reading text, document use, measurement and calculation*);
- E2.5** demonstrate an understanding of and apply the work habits that are important for success in the construction industry, as identified in the Ontario Skills Passport (*e.g., working safely, teamwork, reliability*);
- E2.6** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in construction technology (*e.g., Passport to Safety certificate, technical drawings, reports, photographs of projects, Ontario Skills Passport Work Plan*), and explain why having a current portfolio is important for career development and advancement.